Sta- 230-03 Project (F23)

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```
## Load packages
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyr)
```

Visualization 1

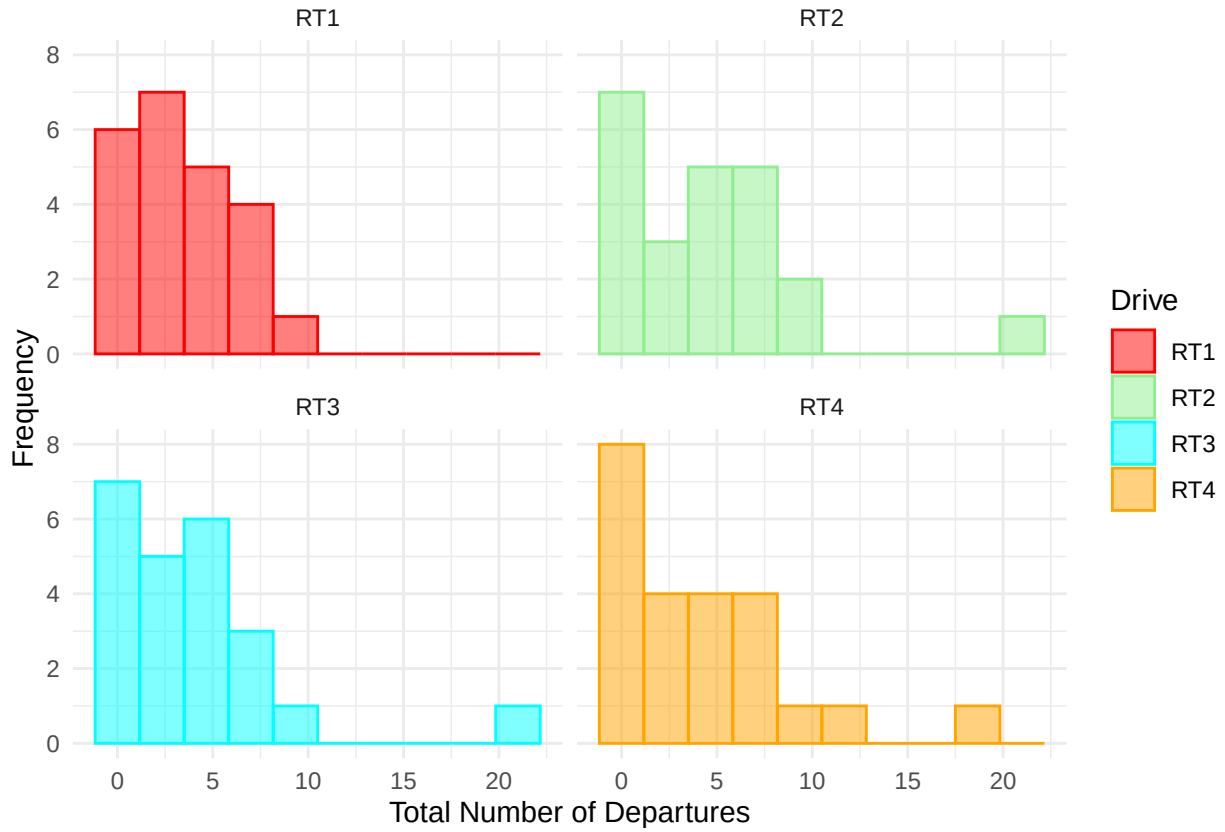
```
drives <- read.csv("https://remiller1450.github.io/data/drives.csv") ## Load data set


drives_NA <- na.omit(drives) %>% ## Remove observations with missing values
  group_by(Subject, Drive) %>%
  mutate(Total_Departures = LaneDeparts_Left +
           LaneDeparts_Right) %>% ## Add column combining left and right departures
  summarize(total = sum(Total_Departures))
```

```
## `summarise()` has grouped output by 'Subject'. You can override using the
## `.groups` argument.
```

```
ggplot(drives_NA, aes(x = total, color = Drive,
                      fill = Drive)) + ## Create the histogram
  geom_histogram(bins = 10, alpha = .5,
                 position = "identity") +
  facet_wrap(~ Drive) + ## Divide into four sub graphs to show each four drives
  labs(
       x = "Total Number of Departures",
       y = "Frequency") + ## Label graph
```

```
  scale_fill_manual(values = c("RT1" = "red", "RT2" = "lightgreen",
                               "RT3" = "cyan", "RT4" = "orange")) +
  scale_color_manual(values = c("RT1" = "red", "RT2" = "lightgreen",
                                "RT3" = "cyan", "RT4" = "orange")) +
  theme_minimal()
```



Visualization 2

```
target <- read.csv("https://collinn.github.io/teaching/2023/data/target.csv")
cohort <- read.csv("https://collinn.github.io/teaching/2023/data/cohort.csv")
## Load data set

target_join <- left_join(x = target, y = cohort, by = c("subjectID", "time")) ## Merge
## data sets "target" and "cohort"
target_longer<- pivot_longer(target_join,
             cols = !subjectID & !time,
             names_to = "Target",
             values_to = "Proportion") ## Rearrange data to combine
## experimental conditions and generate a column
## for the proportion of looks the subject looked at
## "Target" and "Cohort" images
target_separate <- separate(target_longer,
                        col = Target,
```

```
                      into = c("Noise", "Method"),
                      sep = "[.]") ## Separate column into experimental condition
## and type of image
```

```
ggplot(target_separate, aes(x = time, y = Proportion)) + ## Create line graph
  geom_line(aes(color = factor(subjectID))) +
  geom_smooth(method = "loess", color = "black") +
  facet_grid(Noise ~ Method) + ## Two categorical variables are present
## therefore this function is required to display
## 6 sub graphs
  labs(x = "Time", y = "Proportion") +
  theme_bw() +
  theme(legend.position = "none")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

